

Element B Initial Template

Element B: Documentation and Analysis of Prior Design Attempts

Prior Design Attempt #1 - <Shade-a-Rella>



<https://www.shade-a-rella.com/>

Shade-a-rella are shade cloth holders so you can bring the shade to individual plants! Although they are unable to shade multiple plants and they cannot move to give plants more sun manually.

Prior Design Attempt #2 - Garden Shade disk



<https://www.shadeforplants.com/resources.html>

The Garden Shade disk is a stick with a circular shade on the top side of the stick. It blocks the sun from producing too many rays for the plant. As the sun rotates during the day, it hits different angles on the plant.

This design is very successful because when the sun rotates through the day, the angle changes so the sun isn't always on the plant.

Prior Design Attempt #3 - Plant Shade Cloth



<https://www.amazon.com/Greenhouse-Sunblock-Outdoor-Pergola-Chicken/dp/B0BTP5XDKW?th=1>

The Shade Cloth -for Garden Plants,60-70% Sunblock Shade Netting -for Outdoor Garden Netting Mesh Pergola Patio Lawn Sun Shade Cloth. This cloth is able to shade multiple plants but isn't able to move on its own when the plant is getting too much light. This sheet can be placed anywhere in your yard and can be moved very easily. Customers who have bought this shade cloth to cover their plants. Another major problem with this shade cloth is that it is not automatic and has to be moved manually

Prior Design Attempt #4 - <Plant Shade Tent>



<https://www.amazon.com/HFYZZ-Resistant-Sunblock-Protection-Sunlight/dp/B0D4HTDFHK>

The garden shade tent is made of high-quality polyethylene material, wear-resistant, breathable and durable. The shade nets can achieve 80% shading effect while allowing air and moisture to enter, protecting plants from direct sun in hot summer.

Provide evidence and explain what parts of this prior design attempt were unsuccessful at solving the defined problem and why based on design requirements.

Prior Design Attempt #5 - <Garden shade cover >



<https://www.homedepot.com/p/12-ft-x-20-ft-Black-80-Greenhouse-Sunblock-Shade-Cloth-GSDR8012020B/320799018>

- Garden shade cover for plants heat protection and gazebo cooling
- Made of durable UV resistant polyethylene with good privacy
- Easy to install with the supporting sturdy clips